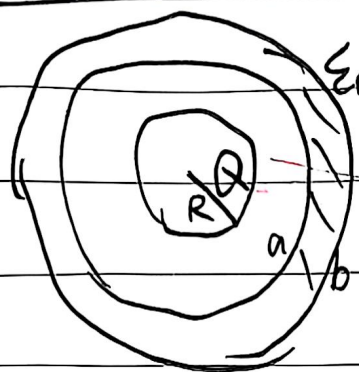


10.19



$E = 0$ for $r < R$

(1) 外: $\vec{E}_{\text{外}} = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2}$ ($R < r < a$ 或 $r > b$)

$\vec{D}_{\text{外}} = \epsilon \vec{E}_{\text{外}} = \epsilon_0 \vec{E}_{\text{外}} = \frac{1}{4\pi} \frac{Q}{r^2}$

内: $\vec{E}_{\text{内}} = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2}$

$\vec{E}_{\text{内}} = \frac{1}{\epsilon_r} \vec{E}_0 = \frac{1}{4\pi\epsilon_0\epsilon_r} \frac{Q}{r^2}$

$\vec{D}_{\text{内}} = \epsilon_0\epsilon_r \vec{E}_{\text{内}} = \frac{1}{4\pi} \frac{Q}{r^2}$ ($a \leq r \leq b$)

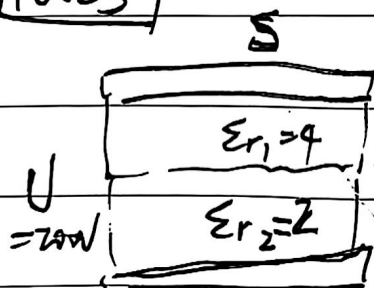
(2) $\vec{P} = (\epsilon_r - 1) \vec{E}_{\text{内}} = \frac{\epsilon_r - 1}{4\pi\epsilon_r} \frac{Q}{r^2}$

$\sigma_{\text{内}} = \vec{P} \cdot \vec{n} = |\vec{P}| = \frac{\epsilon_r - 1}{4\pi\epsilon_r} \frac{Q}{a^2}$ 方向

$\sigma_{\text{外}} = \frac{\epsilon_r - 1}{4\pi\epsilon_r} \frac{Q}{b^2}$

10.23

Attention: $\vec{D}$ 与自由电荷有关, 通过高斯, 应是



$\vec{D}$ 连续, 而非 $\vec{E}_0$ ($\vec{E}$ 在介质

交界处突变) ③ 不能通过 $\frac{U}{d}$ 求初始电场

因为电介质改变了自由电荷的分布!!

(1) $\vec{E}_0 = \frac{U}{d_1 + d_2} = 4 \times 10^4 \text{ V}$

$U = E_1 d_1 + E_2 d_2 = \frac{E_0}{\epsilon_r} d_1 + \frac{E_0}{\epsilon_r} d_2$

(如果两板连接则厚) $E \neq \frac{U}{d}$

$\therefore E_1 = \frac{1}{\epsilon_r} E_0 = 1 \times 10^4 \text{ V}$

$E_2 = \frac{1}{\epsilon_r} E_0 = 2 \times 10^4 \text{ V}$

$\therefore W_1 = \frac{1}{2} \epsilon E_1^2 = \frac{1}{2} \epsilon_0 \epsilon_r E_1^2 = 1.77 \times 10^{-3} \text{ J/m}^3$ [单电介质时 $E = \frac{U}{d}$]

$W_2 = \frac{1}{2} \epsilon_0 \epsilon_r E_2^2 = 3.54 \times 10^{-3} \text{ J/m}^3$ || $U = E_1 d_1 + E_2 d_2 = \frac{D}{\epsilon_r} d_1 + \frac{D}{\epsilon_r} d_2$

(2) $E_1 = W_1 \cdot S \cdot d_1 = 1.42 \times 10^{-8} \text{ J}$

$E_2 = W_2 \cdot S \cdot d_2 = 4.25 \times 10^{-8} \text{ J}$

$\Rightarrow \vec{D} = \epsilon_0 \epsilon_r \vec{E}$

$E_1 = \frac{D}{\epsilon_0 \epsilon_r} = 2.5 \times 10^4$

$E_2 = 5 \times 10^4$

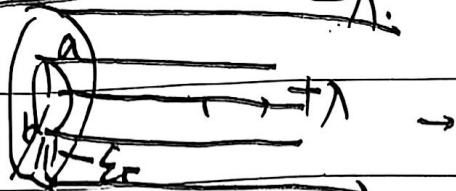
$W_1 = 1.1 \times 10^{-2} \text{ J}$

$W_2 = 2.27 \times 10^{-2} \text{ J}$

(2) $E_1 = W_1 S d_1 = 8.28 \times 10^{-8} \text{ J}$

$E_2 = 2.66 \times 10^{-7} \text{ J}$

10.26



取 Δx , $E_0 = \frac{1}{2\lambda\epsilon_0} \frac{\Delta}{r}$

$E = \frac{1}{\epsilon_r} E_0 = \frac{1}{2\lambda\epsilon_0\epsilon_r} \frac{\lambda}{r}$

$\therefore r$ 处, $w = \frac{1}{2} \epsilon E^2 = \frac{1}{2} \cdot \frac{1}{4\pi^2\epsilon_0\epsilon_r} \frac{\lambda^2}{r^2}$

$\therefore E = \int_a^b w(r) \cdot 2\pi r dr = \frac{\lambda^2}{4\pi\epsilon_0\epsilon_r} \int_a^b \frac{1}{r} dr = \frac{\lambda^2}{4\pi\epsilon_0\epsilon_r} \ln \frac{b}{a}$

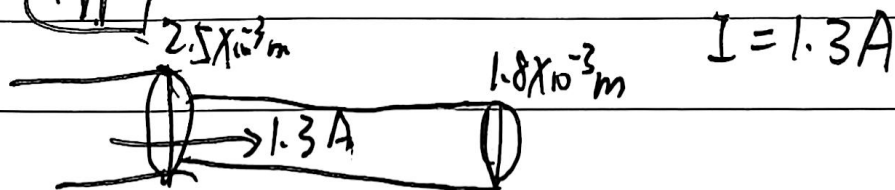
11.2

$I = 1.6 \mu A = neS$

设每秒 n 个电子

$I = ne \Rightarrow n = \frac{9.99}{3.14 \times 10^{-12}} \times 10^{12}$

11.4



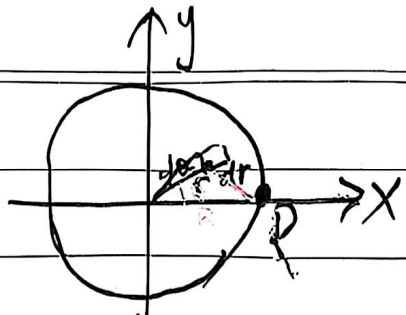
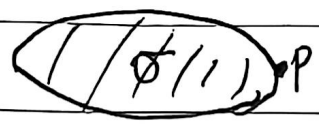
(1) $\vec{j}_1 = \frac{I}{S_1} = \frac{I}{\pi (\frac{d_1}{2})^2} = 2.65 \times 10^5 A/m^2$

$\vec{j}_2 = \frac{I}{S_2} = \frac{I}{\pi (\frac{d_2}{2})^2} = 5.1 \times 10^5 A/m^2$

(2) $\vec{j}_1 = ne v_d$

$n = \frac{\rho_{铜} N_A}{64 \times 10^{-3}} \Rightarrow v_d = \frac{j_1}{\frac{\rho_{铜} N_A}{64 \times 10^{-3}} e} = 0.0196 m/s$

9.34



$\sqrt{R^2 + r^2 - 2Rr \cos \theta}$ (不好算)

$$d\varphi = \frac{1}{4\pi\epsilon_0} \frac{dq}{r} = \frac{1}{4\pi\epsilon_0} \frac{\sigma \cdot r dr d\theta}{r}$$

$$= \frac{\sigma}{4\pi\epsilon_0} dr d\theta$$

$$\therefore \varphi = \int_0^{2\pi} \int_0^R d\varphi = \frac{\sigma}{4\pi\epsilon_0} \cdot R \cdot 2\pi = \frac{\sigma R}{2\epsilon_0}$$

取 $dQ = \sigma \cdot 2\pi r dr$

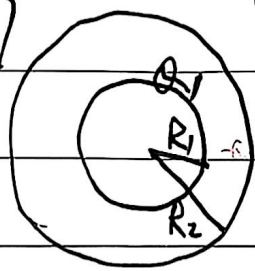
$$d\varphi = \frac{1}{4\pi\epsilon_0} \frac{\sigma \cdot 2\pi r dr}{R}$$

$$\frac{\sigma dr}{2\epsilon_0 R}$$

$$\varphi = \int_0^R \frac{\sigma dr}{2\epsilon_0 R}$$

$$= \frac{\sigma R}{2\epsilon_0}$$

10.7



(1) $\varphi_A = \frac{1}{4\pi\epsilon_0} \frac{Q_1}{R_1} + \frac{1}{4\pi\epsilon_0} \frac{Q_2}{R_2} = 0$

$$\Rightarrow Q_2 = -\frac{Q_1}{R_1} R_2$$

(2)
$$\begin{cases} 0 & 0 < r < R_1 \\ \varphi = \frac{1}{4\pi\epsilon_0} \frac{Q_1 + Q_2}{r} & r > R_2 \end{cases}$$

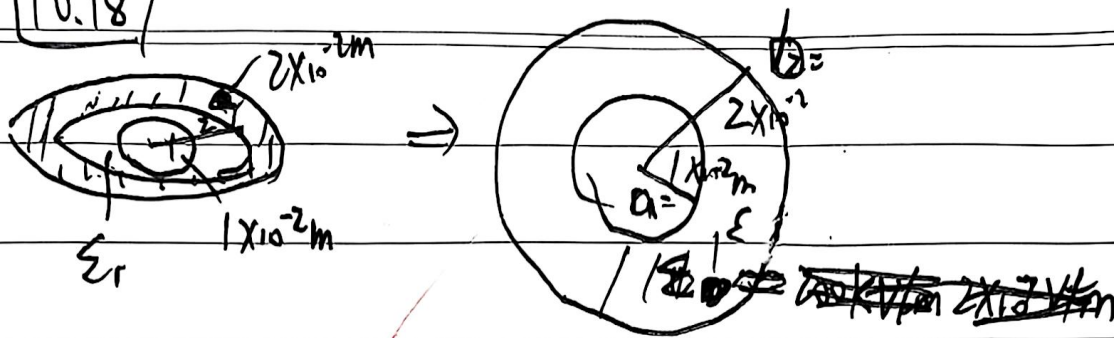
当 $R_1 < r < R_2$, 高斯: $E = \frac{1}{4\pi\epsilon_0} \frac{Q_1}{r^2}$

$$\therefore \varphi - 0 = \int_R^r E dr = \frac{Q_1}{4\pi\epsilon_0} \left(\frac{1}{R_1} - \frac{1}{r} \right)$$

$$\frac{R_1 - R_2}{R_1} Q_1$$

$$\begin{cases} 0 & 0 < r < R_1 \\ \frac{Q_1}{4\pi\epsilon_0} \left(\frac{1}{R_1} - \frac{1}{r} \right) & R_1 < r < R_2 \\ \frac{Q_1}{4\pi\epsilon_0} \left(-\frac{R_1 - R_2}{R_1 r} \right) & r > R_2 \end{cases}$$

10.18



$$E_{\max} = 200 \times 10^3 / 10^{-2} \text{ V} = 2 \times 10^7 \text{ V}$$

由一电场强度 = 最大电场强度 = $2 \times 10^7 \text{ V/m}$

设 V 下求 $E_{\max}$

$$\text{电场分布: } \frac{Q}{\epsilon_0} = 2\pi r k E \Rightarrow E = \frac{k}{r}$$

$$\text{当 } r = a \text{ 时, } E_{\max} = \frac{k}{a} = 2 \times 10^7 \text{ V/m} \Rightarrow k = 2 \times 10^7 a$$

$$V_{\max} = \int_a^b \frac{k}{r} dr = k \ln \frac{b}{a} = 139 \text{ kV}$$

10.28

σS_{++} 求 W_1 :



设 σ , $\frac{\sigma \cdot \Delta S}{\epsilon_0} = E \cdot \Delta S \Rightarrow \sigma = \epsilon_0 E = \epsilon_0 \frac{U}{d-t}$

$\therefore C = \frac{Q}{U} = \frac{\sigma \cdot S}{U} = \frac{\epsilon_0}{d-t} S = 1.33 \times 10^{-10} \text{ F}$

$W_1 = \frac{1}{2} C U^2 = 2.39 \times 10^{-5} \text{ J}$

$Q = C U = 7.98 \times 10^{-8} \text{ C}$

② 求 W_2 : $C' = \epsilon_0 \frac{S}{d} = 8.85 \times 10^{-11} \text{ F}$

$\therefore W_2 = \frac{1}{2} C' U^2 = 3.60 \times 10^{-5} \text{ J}$

$\therefore \Delta W = W_2 - W_1 = 1.21 \times 10^{-5} \text{ J}$

(2) $E = \frac{U}{d} = \frac{Q}{C' d} = 3 \times 10^5 \text{ V/m}$

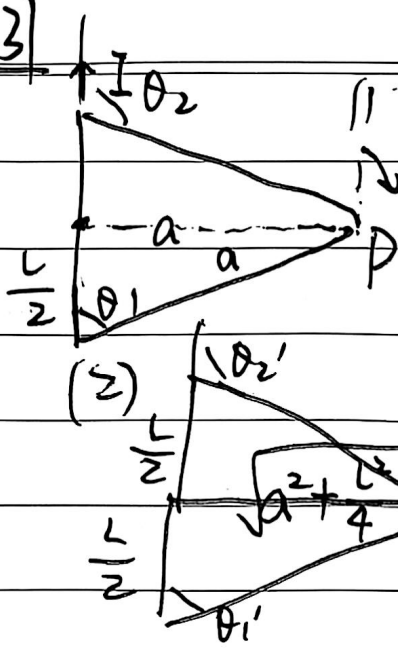
$\therefore F = Q E = 0.024 \text{ N}$

0.012

$\frac{Q}{\epsilon_0 S d}$
不应取合场强

应取单极板产生场 $\frac{\sigma}{2\epsilon_0}$

12.3



$$\cos \theta_1 = \frac{\frac{L}{2}}{\sqrt{\frac{L^2}{4} + a^2}}$$

$$\begin{aligned} \text{1) } \vec{B} &= \frac{\mu_0 I}{4\pi a} (\cos \theta_1 - \cos \theta_2) \\ &= \frac{\mu_0 I}{4\pi a} \frac{L}{\sqrt{\frac{L^2}{4} + a^2}} \end{aligned}$$

垂直纸面向下

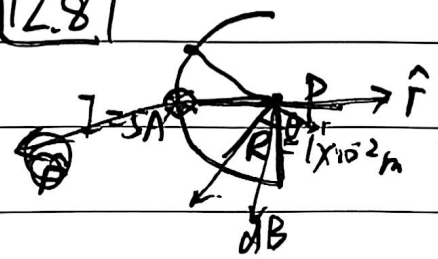
[理解成垂直纸面的正上方了]
[这里指平面内]

$$B' = \frac{\mu_0 I}{4\pi \sqrt{a^2 + \frac{L^2}{4}}} \frac{L}{\sqrt{\frac{L^2}{4} + a^2}}$$

$\frac{\mu_0 I L}{4\pi a \sqrt{L^2 + 4a^2}}$

$$R = \rho \frac{L}{S} \quad I \propto S$$

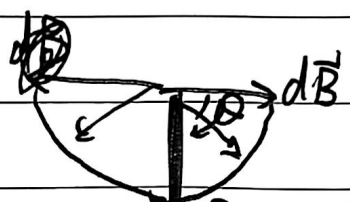
12.8



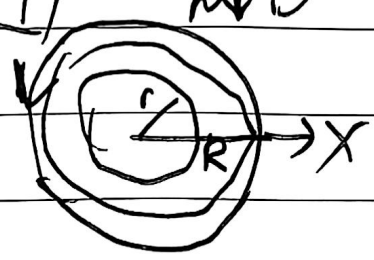
$$dB = \frac{\mu_0 I}{2\pi R} d\theta$$

$$\begin{aligned} B &= \int_0^\pi dB \sin \theta = \frac{\mu_0 I}{2\pi^2 R} \int_0^\pi \sin \theta d\theta \\ &= \frac{\mu_0 I}{\pi^2 R} = 6.37 \times 10^{-5} \text{ T} \end{aligned}$$

方向如图



12.9

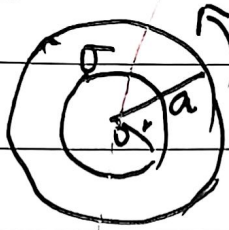


当半径为x时, $dB = \frac{\mu_0 I}{2x} \cdot \frac{dx}{R-r} N$

$$B = \int dB = \int_r^R \frac{\mu_0 I N}{2(R-r)} \frac{dx}{x} = \frac{\mu_0 I N}{2(R-r)} \ln \frac{R}{r}$$

垂直纸面向外

12.11



$$dI = dq \cdot v = dq \frac{w}{2\pi} = \sigma w r dr$$

r 处取圆环 $dq = \sigma \cdot 2\pi r dr$

$$dB = \frac{\mu_0 dq w r}{2r} = \frac{\mu_0 \sigma w}{2} r dr$$

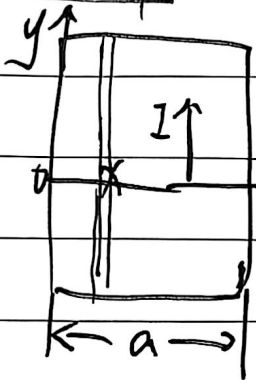
$$\therefore B = \int_0^a dB = \frac{\mu_0 \sigma w}{2} \cdot \frac{a^2}{2}$$

轴上一点 $dB = \frac{\mu_0 \sigma w r dr}{2(r^2 + x^2)^{3/2}}$ $B = \int_0^a dB = \frac{\mu_0 \sigma w}{2} \left[\frac{2x^2 + a^2}{x^2 + a^2} - 2x \right]$

x 处取无限长直导线

$$dI = \frac{I}{a} dx \quad dB = \frac{\mu_0 dI}{2\pi(2a-x)} = \frac{\mu_0 I dx}{2\pi a(2a-x)}$$

12.13



$$\therefore B = \int dB = \int_0^a \frac{\mu_0 I}{2\pi a} \frac{dx}{2a-x} = \frac{\mu_0 I}{2\pi a} \ln 2$$